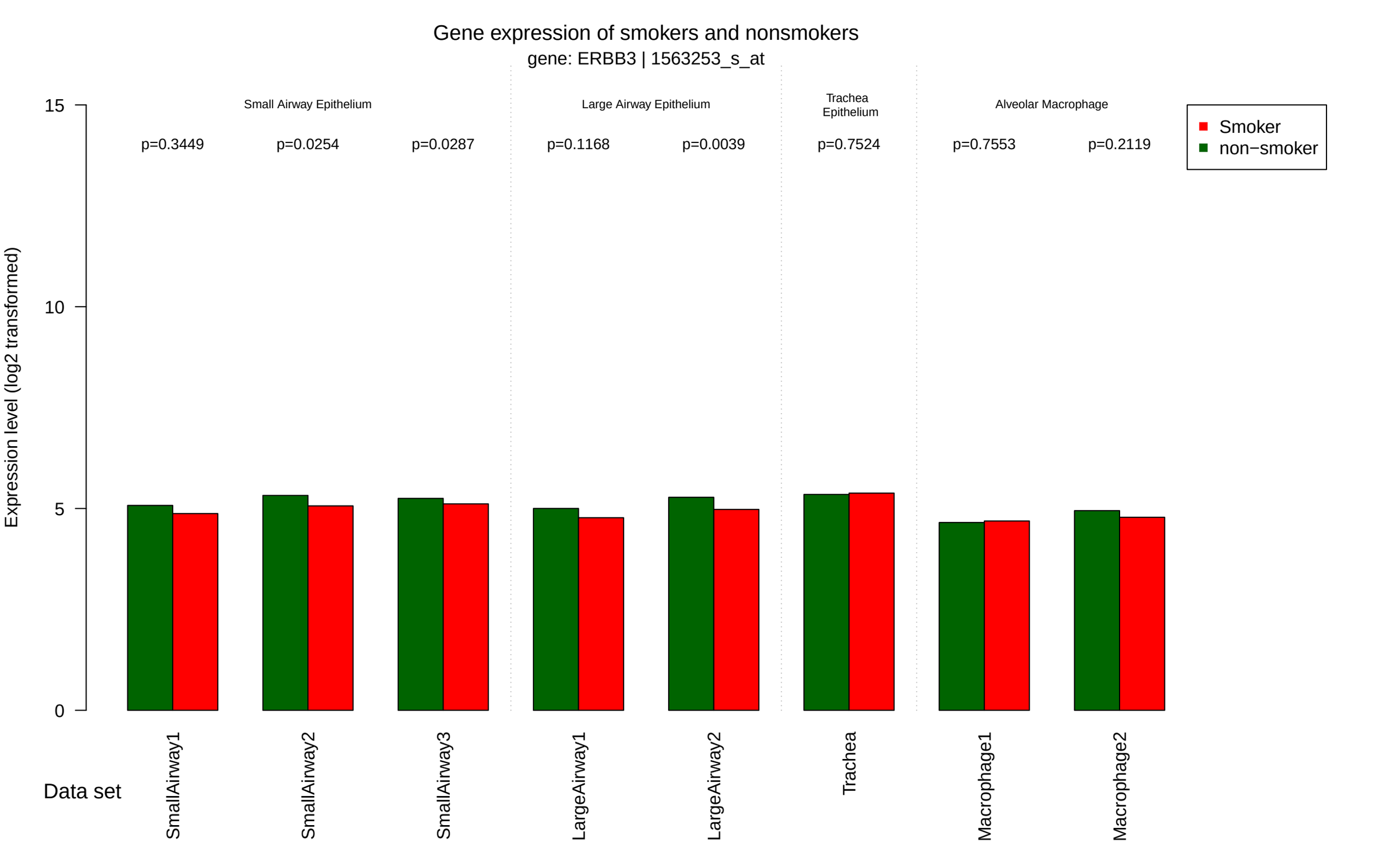
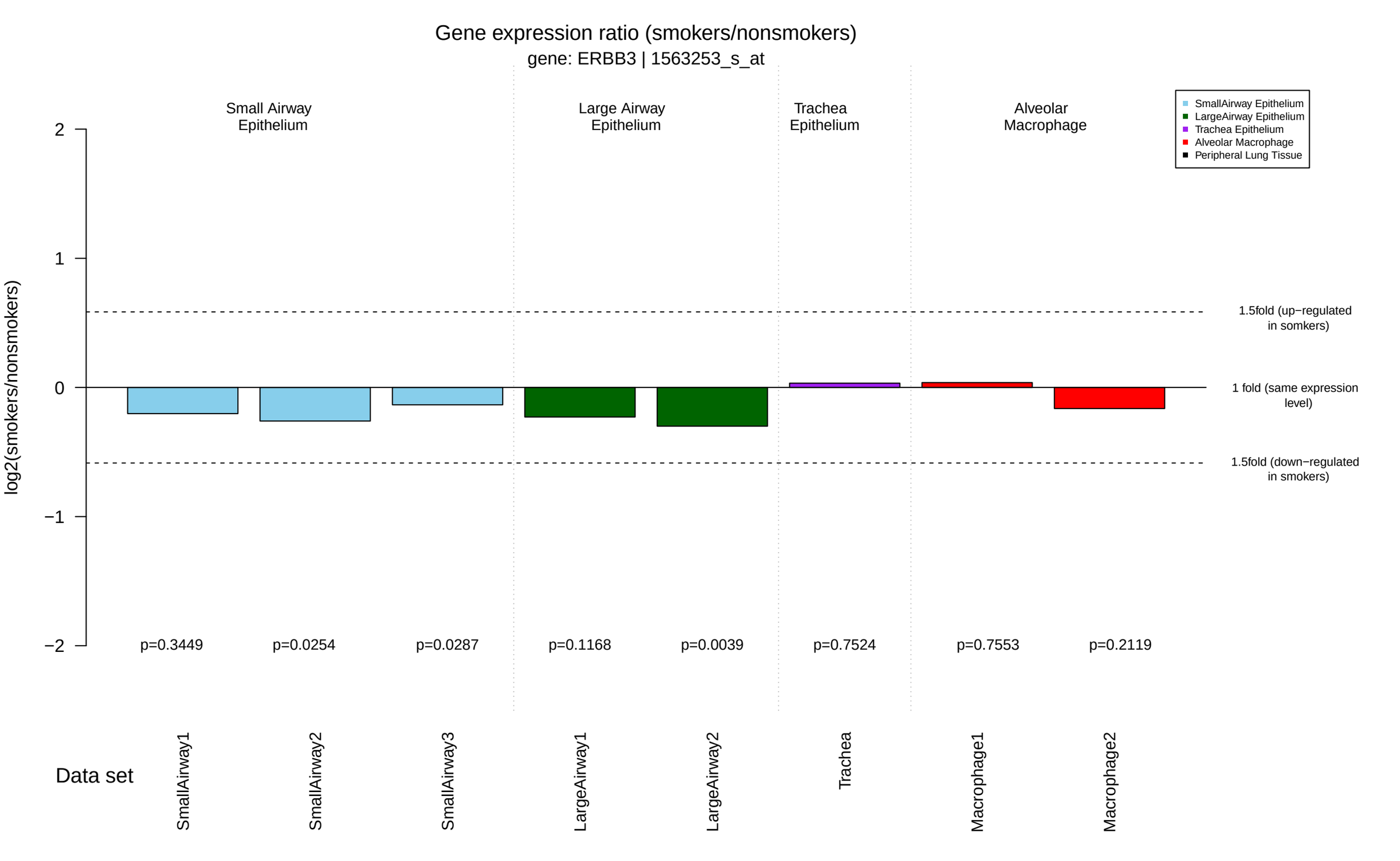
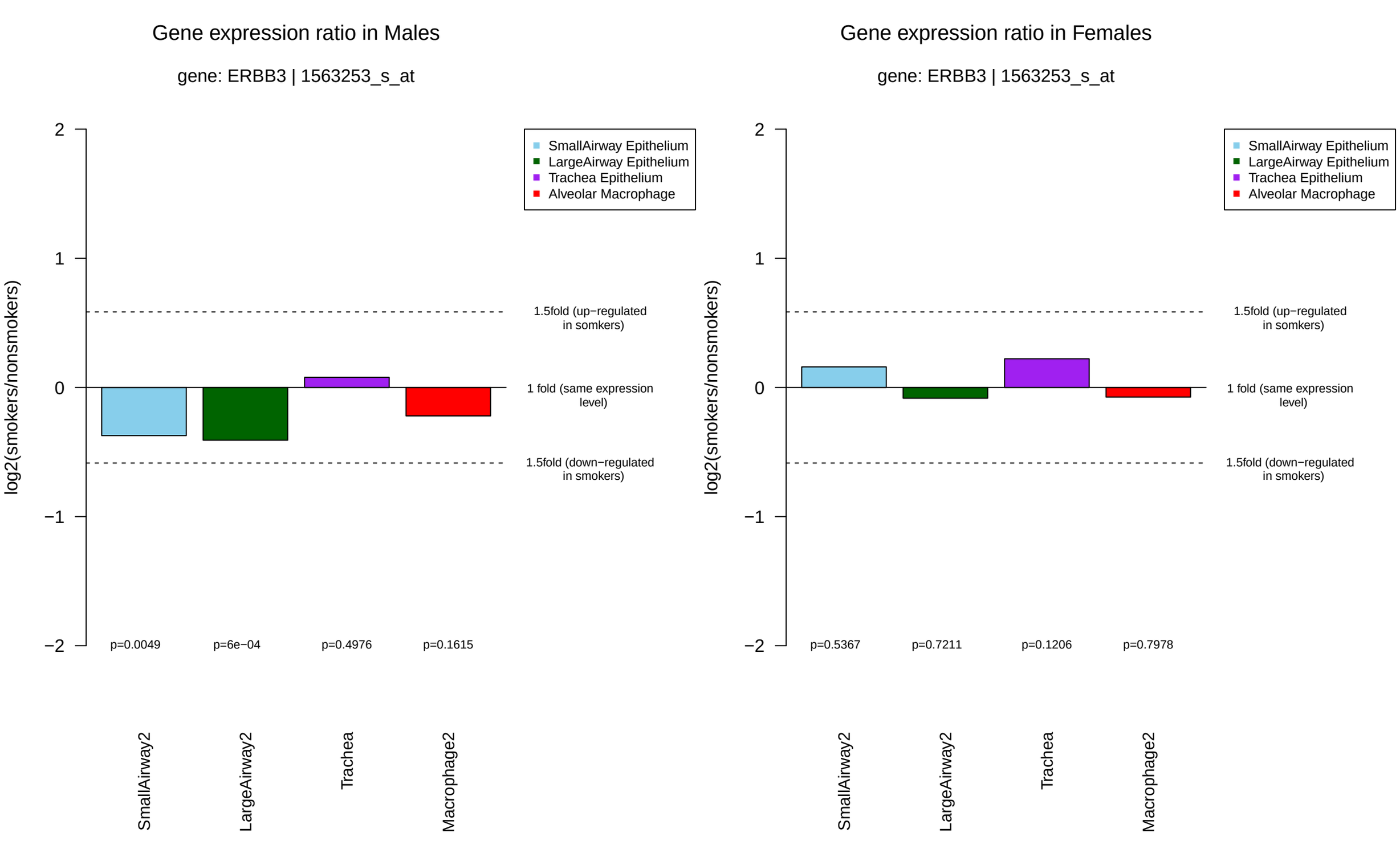
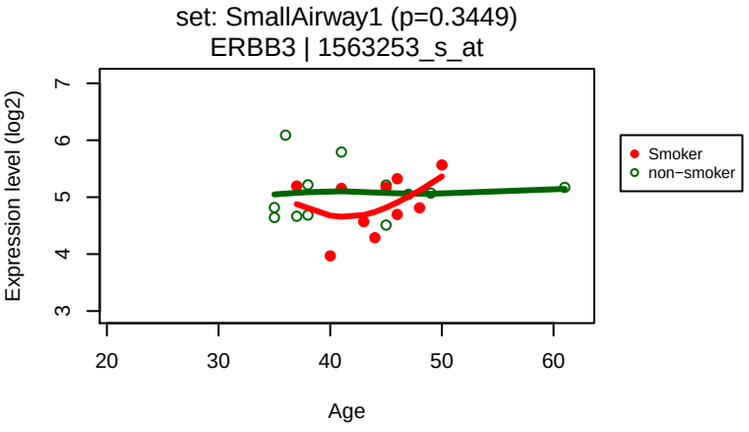
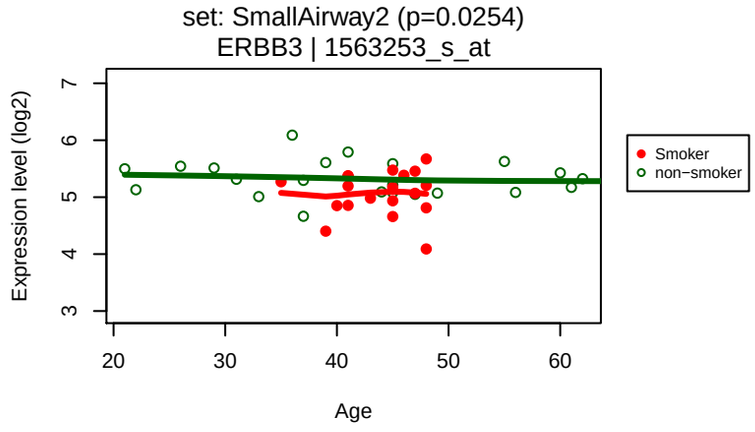
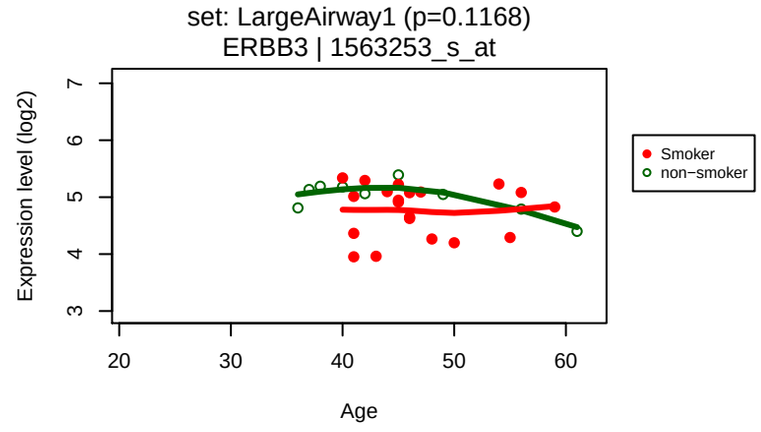

Gene expression of smokers and nonsmokers
gene: ERBB3 | 1563253_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3449
p=0.0254
p=0.0287
p=0.1168
p=0.0039
p=0.7524
p=0.7553
p=0.2119
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

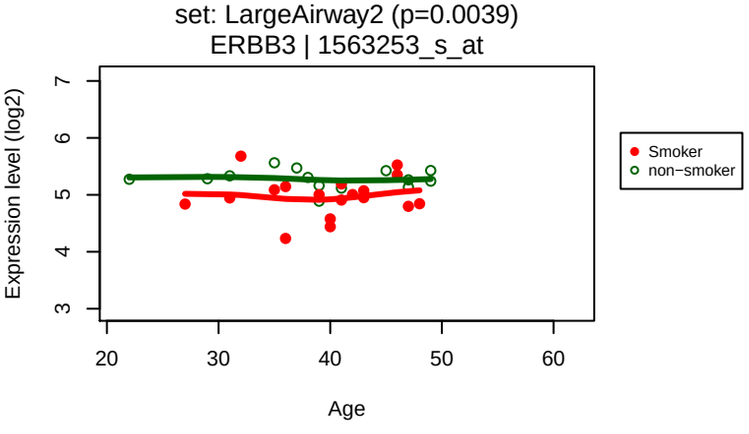

Gene expression ratio (smokers/nonsmokers)
gene: ERBB3 | 1563253_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3449
p=0.0254
p=0.0287
p=0.1168
p=0.0039
p=0.7524
p=0.7553
p=0.2119
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

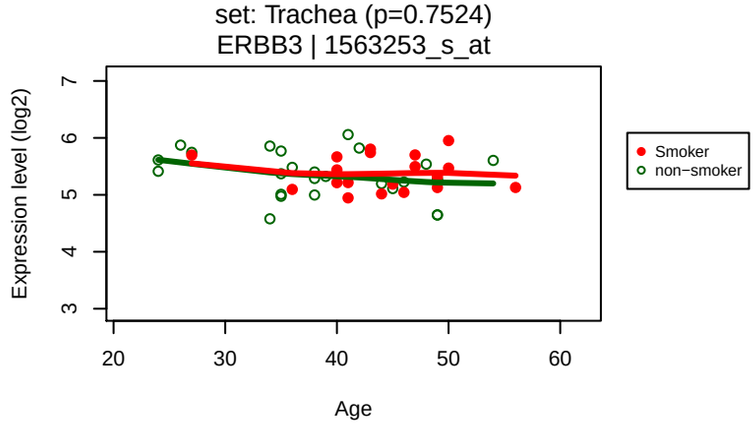

Gene expression ratio in Males
gene: ERBB3 | 1563253_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0049
p=6e−04
p=0.4976
p=0.1615
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ERBB3 | 1563253_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5367
p=0.7211
p=0.1206
p=0.7978
SmallAirway2
LargeAirway2
Trachea
Macrophage2

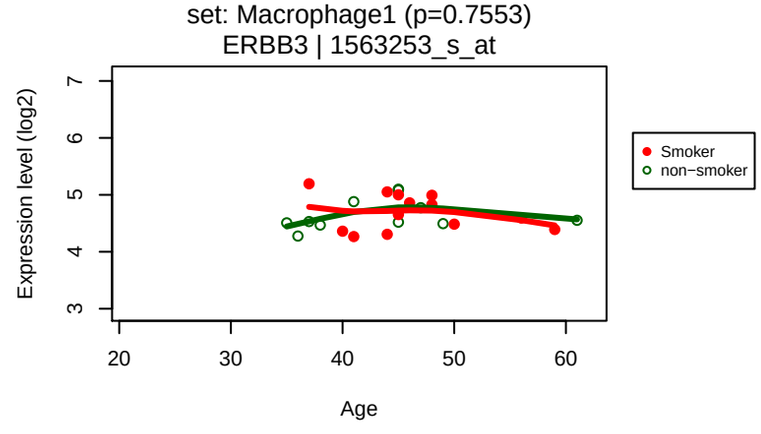

set: SmallAirway1 (p=0.3449)
ERBB3 | 1563253_s_at
Expression level (log2)
Age
Smoker
non−smoker

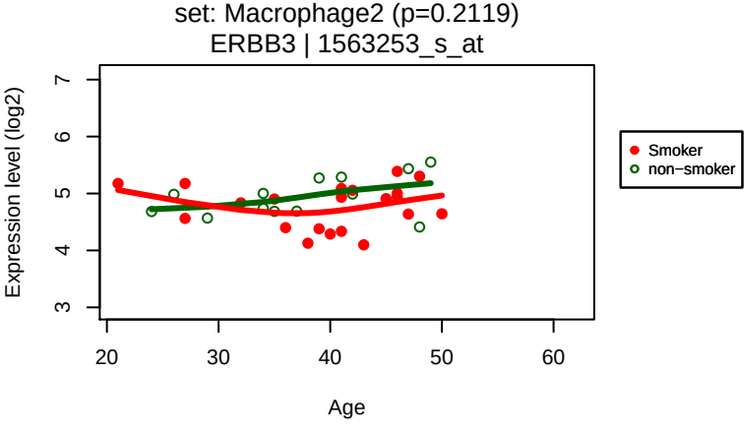

set: SmallAirway2 (p=0.0254)
ERBB3 | 1563253_s_at
Expression level (log2)
Age
Smoker
non−smoker

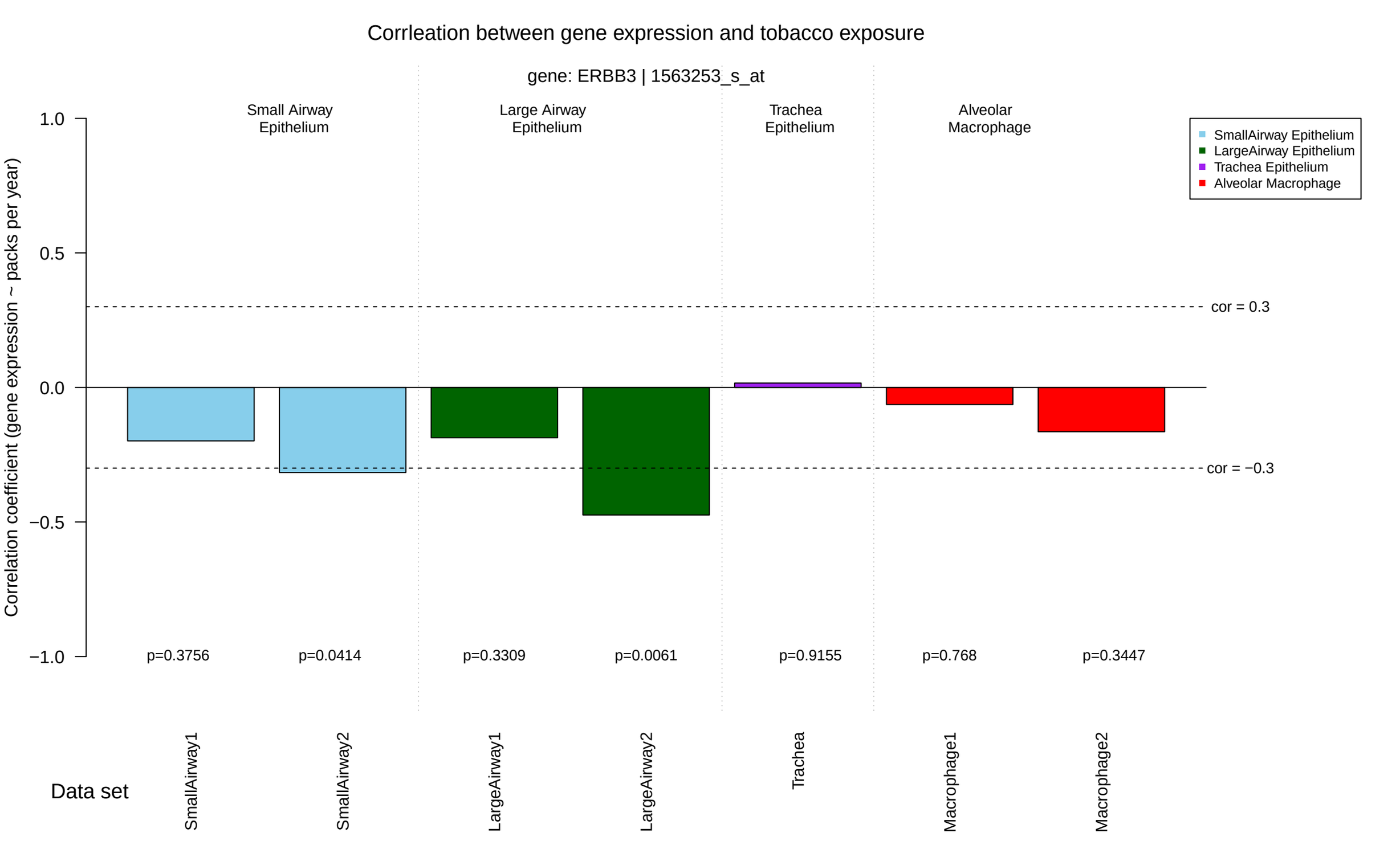

set: LargeAirway1 (p=0.1168)
ERBB3 | 1563253_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0039)
ERBB3 | 1563253_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7524)
ERBB3 | 1563253_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7553)
ERBB3 | 1563253_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2119)
ERBB3 | 1563253_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ERBB3 | 1563253_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3756
p=0.0414
p=0.3309
p=0.0061
p=0.9155
p=0.768
p=0.3447
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2